

Vector Search Algorithms and Graph Indexing

HNSW Graph Indexing

Hierarchical Navigable Small World (HNSW) graphs construct multi-layer proximity graphs for approximate nearest neighbor (ANN) search. HNSW achieves logarithmic search time complexity $O(\log N)$ while maintaining over 95% recall on high-dimensional vector spaces.

Distance Metrics Comparison

Cosine similarity measures vector angle regardless of magnitude, Euclidean distance computes L2 geometric distance, and Dot Product evaluates unnormalized projection. Normalized vectors make Dot Product identical to Cosine similarity for BLAS acceleration.

Quantization Techniques

Product Quantization (PQ) and Scalar Quantization (SQ8) compress 32-bit floating-point embeddings into 8-bit or 4-bit codes, reducing RAM consumption by 75% with minimal accuracy degradation.